

SARGASSUM ALGAL BLOOMS AND HOUSEHOLD WELFARE IN THE DOMINICAN REPUBLIC

ABSTRACT

This paper estimates the effect of Sargassum seaweed inundation on household income in the Dominican Republic using a monthly municipality-level panel from 2017 to 2025. Combining satellite data on Sargassum intensity with quarterly household survey data from 100 municipalities, I find that Sargassum significantly reduces household income during peak bloom months (May–September), with an estimated effect of -73.90 log points ($p < 0.01$). The effect is largest in non-coastal municipalities, suggesting disruption to tourism supply chains extends well beyond the shoreline. Among income groups, the negative effect is largest for the poorest households and becomes statistically detectable for low-income earners when zero-income households are excluded from the analysis. Results are identified through municipality and year-by-month fixed effects, which control for time-invariant geographic differences and all common national shocks.

1. Introduction

The Dominican Republic is among the countries most severely affected by the mass influx of pelagic Sargassum seaweed that has intensified across the tropical Atlantic since approximately 2011. In years such as 2018, 2021, 2022, 2023, and 2025, enormous blooms washed ashore along the country's Caribbean and Atlantic coastlines, fouling beaches, generating toxic hydrogen sulphide gas as the seaweed decomposed, and disrupting the marine environments on which coastal communities depend. For a country where tourism accounts for roughly 8% of GDP and attracts around eight million visitors annually — including the internationally known Punta Cana resort corridor — repeated Sargassum inundation represents a sustained economic threat.

Despite growing ecological and public health documentation of this phenomenon, systematic evidence on its economic consequences at the household level remains scarce. This paper helps fill that gap by estimating the effect of Sargassum intensity on household income across Dominican municipalities from 2017 to 2025.

I combine two data sources. The first is satellite-derived monthly Sargassum intensity data constructed from NASA's MODIS-Aqua sensor using the Alternative Floating Algae Index (AFAI). The second is the Encuesta Nacional Continua de Fuerza de Trabajo (ENCFT), the Dominican Republic's quarterly household labour force and income survey. Merging these at the municipality-month level produces a panel of 10,263 observations across 100 municipalities. I estimate the income effect of Sargassum using OLS regression with municipality and year-by-month fixed effects, which absorb permanent geographic differences and all shocks common to all municipalities in a given month.

Three findings emerge. First, Sargassum significantly reduces household income during peak bloom months (May–September), but the effect is not detectable when averaged across the full year, suggesting the harm is concentrated in the months when blooms are actually present. Second, the peak-season effect is paradoxically stronger in non-coastal municipalities than coastal ones, pointing to disruption of tourism supply chains and regional labour markets that extend beyond the immediate shoreline. Third, the income loss follows the expected distributional pattern — largest for poorer households — though this gradient only becomes statistically clear for the bottom group when households reporting zero income are excluded.

The remainder of the paper proceeds as follows. Section 2 provides background on Sargassum. Section 3 describes the data. Section 4 presents the empirical strategy. Section 5 reports results. Section 6 concludes.

2. Background

Sargassum is a genus of brown macroalgae that forms vast floating mats across the tropical Atlantic. What researchers now call the Great Atlantic Sargassum Belt stretches from the west coast of Africa to the Gulf of Mexico and the Caribbean. While Sargassum has always existed in the Atlantic in moderate quantities — providing important habitat for marine life — the volume of pelagic Sargassum has increased dramatically since approximately 2011. Scientists attribute this to a combination of elevated ocean surface temperatures, shifting Atlantic circulation patterns, and increased nutrient runoff from deforestation and intensive agriculture in West Africa and the Amazon basin.

When Sargassum washes ashore in large quantities, it decomposes and releases hydrogen sulphide and ammonia, gases that are hazardous to human health at high concentrations. Thick mats of rotting seaweed make beaches inaccessible for tourism, entangle fishing gear, degrade water quality, and can damage coral reef ecosystems. For Caribbean tourism economies, the visual and olfactory impact is particularly damaging: a beach covered in dark, decomposing seaweed cannot attract the international visitors that coastal communities depend on.

The Dominican Republic's exposure is acute. Its coastline borders both the Caribbean Sea and the Atlantic Ocean, and the prevailing currents and winds bring Sargassum directly to its shores during the peak season of May through September. While some coastal communities have developed rudimentary collection and removal operations, the scale of recent blooms has repeatedly overwhelmed local capacity.

3. Data

3.1 Satellite Sargassum Data

I measure Sargassum exposure using the Alternative Floating Algae Index (AFAI), a spectral index derived from NASA's MODIS-Aqua satellite that detects floating vegetation at the ocean surface. Monthly AFAI data at 4km spatial resolution are sourced from NASA's ERDDAP data server, covering the waters within 20km of the Dominican coastline from January 2017 to December 2025.

For each municipality-month observation, Sargassum intensity is computed as the mean value of $\max(\text{AFAI} - 0.001, 0)$, where the 0.001 threshold removes background noise. This measure captures both the spatial coverage and density of Sargassum mats in the coastal waters adjacent to each municipality.

Municipalities are classified as coastal (48 of 100 in the analysis sample) if their administrative boundaries directly touch the ocean shoreline. The Dominican Republic–Haiti land border is excluded from this classification to avoid incorrectly treating border municipalities as coastal. Non-coastal municipalities are assigned Sargassum exposure based on ocean pixels within a 20km buffer, capturing regional offshore exposure rather than direct shoreline contact.

3.2 Household Survey Data

Income data come from the ENCFT, the Dominican Republic's quarterly continuous labour force survey, which is designed to be representative at the quarterly level. I use survey waves from 2016 to 2025. For each municipality-quarter, I compute mean household log income and mean log income separately for each national income tertile group — the bottom, middle, and top thirds of the per-capita household income distribution within each year. These tertile-specific income measures allow me to test whether the Sargassum effect is larger for poorer households.

Approximately 17.6% of surveyed households report zero income. I construct a second set of tertiles that excludes these households and use it as a robustness check, as discussed in Section 5.

3.3 Analysis Panel

I merge monthly satellite data with quarterly survey data by assigning each month the income observation from the survey quarter that contains it. The final panel contains 10,263 observations across 100 municipalities and the years 2017–2025.

4. Empirical Strategy

The main estimating equation is:

$$\log(\text{income}_{it}) = \beta \cdot \text{AFAI}_{it} + \alpha_i + \gamma_t + \varepsilon_{it} \tag{1}$$

where i indexes municipalities and t indexes year-month periods. AFAI_{it} is Sargassum intensity in municipality i in period t . α_i are municipality fixed effects, γ_t are year-by-month fixed effects, and standard errors are clustered at the municipality level.

The municipality fixed effects α_i absorb all time-invariant differences across municipalities — permanent income differences, geographic location, distance from major tourism centres, and baseline economic structure. The year-by-month fixed effects γ_t absorb every shock that affects all municipalities simultaneously in a given calendar month of a given year, including the seasonal pattern of Sargassum itself, national economic trends, and common weather events. After controlling for both sets of fixed effects, identification relies entirely on within-municipality variation in Sargassum intensity over time — months in which a specific municipality experienced more Sargassum than was typical for that place at that time of year.

To test whether the effect is concentrated during peak bloom months, I estimate equation (1) on a subsample restricted to May through September:

$$\log(\text{income}_{it}) = \beta_{\text{peak}} \cdot \text{AFAI}_{it} + \alpha_i + \gamma_t + \varepsilon_{it} \tag{2}$$

To test for heterogeneous effects by income group, I replace the dependent variable with mean log income for each tertile group k :

$$\log(\text{income}_{TK,it}) = \beta_k \cdot \text{AFAI}_{it} + \alpha_i + \gamma_t + \varepsilon_{it} \quad k \in \{\text{bottom}, \text{middle}, \text{top}\} \tag{3}$$

I also estimate equations (1) and (2) separately for coastal and non-coastal municipalities to test whether direct shoreline exposure produces a larger effect than regional spillovers.

The key identifying assumption is that, conditional on municipality and year-by-month fixed effects, no unobserved factor simultaneously drives Sargassum intensity in a specific municipality and that municipality's income within the same month. The most plausible threat would be a concurrent local shock — such as a hurricane — that affects both. I cannot formally rule this out, which is a limitation of the analysis.

5. Results

5.1 Main Results

Table 1 presents the main estimates. The full-year coefficient on Sargassum intensity is negative (-28.15) but statistically insignificant ($p = 0.13$), suggesting a negative relationship that is too noisy to distinguish from zero when averaged across all months including those when Sargassum is largely absent.

The peak season result is markedly sharper. Restricting the sample to May–September yields a coefficient of -73.90 ($p < 0.01$). A one-unit increase in mean excess AFAI during peak months is associated with a 73.90 log-point reduction in municipal household income. The contrast between the full-year and peak-season estimates is consistent with Sargassum being the mechanism: the income loss materialises specifically in the months when blooms are present, not year-round.

5.2 Heterogeneity by Coastal Proximity

Table 3 presents results for coastal and non-coastal municipalities separately. Over the full year, neither group shows a significant effect. During peak season, the significant effect is found in non-coastal municipalities (-80.08 , $p < 0.01$), while coastal municipalities show a negative but statistically insignificant estimate (-58.95).

This finding is initially counterintuitive. Several explanations are plausible. Coastal households tend to have more diversified income sources, combining fishing, tourism employment, and local trade, which may allow partial offsetting of Sargassum shocks when one income source is disrupted. Non-coastal municipalities that supply tourism services — accommodation, transport, food production — may face a large demand-side collapse when Sargassum closes beaches, with fewer alternative income streams. Whatever the mechanism, the finding suggests that the economic costs of Sargassum extend substantially beyond the municipalities that physically border the sea.

5.3 Heterogeneity by Income Tertile

Table 2 presents results by income tertile. Full-year coefficients follow a monotonically decreasing pattern in absolute magnitude: bottom tertile (-144.87), middle tertile (-30.75 , $p < 0.05$), top tertile (-12.20). This gradient is consistent with lower-income households having less ability to insure against environmental shocks through savings, credit access, or income diversification.

Only the middle tertile reaches significance over the full year. The bottom tertile's coefficient, though large, is extremely imprecise ($SE = 90.06$). This occurs for a structural reason: approximately 5% of municipality-month observations in the bottom tertile group report zero household income. Log-transforming these observations produces an outcome of zero regardless of Sargassum intensity, which floors the dependent variable and inflates noise. The standard deviation of log income in the bottom tertile is 1.93 compared to 0.34 for the middle tertile, reflecting this problem directly.

During peak season (Table 2, columns 4–6), the same gradient holds: bottom (-75.07), middle (-48.99 , $p < 0.01$), top (-26.64). The middle tertile peak-season estimate is larger in absolute value than its full-year counterpart (-48.99 vs -30.75), consistent with the effect being concentrated in bloom months.

5.4 Robustness: Excluding Zero-Income Households

Table 4 addresses the zero-income floor problem by excluding the 17.6% of surveyed households that report zero income and recomputing tertile cutoffs on the remaining positive-income sample. The bottom tertile in this sample therefore represents the lowest-earning workers in the Dominican Republic — not households with no income — and has a higher income threshold: for example, the bottom/middle boundary rises from RD\$5,000 to RD\$7,000 per capita per month in 2024.

The results change in an informative way. Full-year coefficients for all three tertile groups are negative but insignificant. During peak season, however, the bottom tertile produces a negative and significant coefficient (-48.69 , $p < 0.01$), and the gradient is monotonically decreasing: bottom (-48.69) $>$ middle (-27.02) $>$ top (-17.69). The significance of the bottom group in the non-zero sample — compared to its insignificance in the original sample — confirms that the zero-income floor was masking a real effect on low-income earners. The poorest workers do experience significant income losses during peak Sargassum months; it is simply not possible to detect this through the income measure for households that report no income at all.

6. Conclusion

This paper provides evidence that Sargassum inundation reduces household income in the Dominican Republic. The main finding is a peak-season income effect of -73.90 log points ($p < 0.01$), identified through within-municipality, within-time variation in Sargassum intensity after controlling for municipality and year-by-month fixed effects.

Two findings extend beyond this headline result. First, the significant peak-season impact on income is concentrated in non-coastal municipalities, suggesting that Sargassum disrupts regional economies through tourism supply chains rather than — or in addition to — direct coastal impacts. This implies that policy responses focused solely on coastal municipalities may underestimate the true economic scope of the problem. Second, the income effect is largest for the poorest households, but only becomes detectable for the bottom group when zero-income households are excluded. This highlights a broader measurement challenge: income-based welfare measures cannot capture the impact on households with no formal income, who may be the most vulnerable to environmental shocks operating through non-income channels such as food security or health costs.

Limitations of this analysis include the use of quarterly rather than monthly income data, the municipality-level aggregation of both treatment and outcome variables, and the inability to formally rule out time-varying local confounders. Future work with finer spatial or temporal resolution, health and nutrition outcomes, or a valid instrumental variable could strengthen identification and provide a more complete picture of Sargassum's welfare consequences in the Caribbean.